



Causal relationship discovery for highway crash analysis using semi-data-driven Bayesian network

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ABSTRACT

With the widespread application of advanced machine learning techniques, researchers need a more transparent decision-making process. The data-driven causal relationship discovery techniques often lack interpretability. Therefore, a semi-data-driven Bayesian network structure learning algorithm, the Expert Knowledge Constraint-based (EKC) algorithm, is proposed. By integrating expert knowledge with conditional independence tests, the EKC algorithm constructs a causal Bayesian network with a high level of interpretability. The algorithm was applied to a highway safety scene using crash data collected in 2022 from the HuNing Highway in China. The effects of the Bayesian network on variables were estimated using the Bayesian estimation algorithm, and the most dangerous scenarios were ranked using the variable elimination algorithm. Key findings include: (1) date-related variables do not directly affect crashes; (2) unfavorable temperatures, medium-level traffic volumes, and snowy weather conditions are associated with higher crash probabilities; and (3) the highest crash probability occurs under medium traffic volume, cold temperatures, winter season, cloudy weather, morning hours, and weekdays. The EKC algorithm was compared with the Hill Climbing algorithm, Chow-Liu Trees algorithm, and logistic model, demonstrating significant improvements in interpretability while maintaining good fitting scores. Furthermore, the definition framework of model interpretability in traffic crash analytics was discussed, including causality, trust, heterogeneity, transferability, and stability.

1. Introduction

While data-driven techniques can handle large-scale datasets and achieve high prediction accuracy, the decision procedure is less interpretable, which is called the black-box nature (Mannering et al., 2020). The need for interpretability makes many Machine Learning (ML) techniques less suitable for digging crash causation. A reliable crash causation chain with interpretability can help managers apply targeted interventions in essential nodes, which can lower costs and be more effective in decreasing crashes. Bayesian Networks (BNs) offer a more interpretable alternative (Liu et al., 2005). BNs are constructed based on Directed Acyclic Graphs (DAGs), which can reveal causal relationships, estimate variable effects, and make inferences. Data-driven BNs have been applied in traffic crash analysis, with Hossain and Muromachi (2012) and Prati et al. (2017) identifying key factors before constructing BNs for prediction. Some arcs in BNs, however, are difficult to explain or even contradict common sense.

This study aims to discover the causal relationship using BNs. A semi-data-driven BN structure learning technique, the Expert Knowledge Constraint-based Algorithm (EKC), is proposed. A typical highway crash scenario was selected, with crash and traffic volume data collected from the Suzhou Road Traffic Crash Analysis and Warning System in 2022 for the HuNing Highway. Beyond structure learning, the BN framework includes parameter learning and inference to estimate causal effects and identify dangerous scenarios. Variable elimination and Bayesian estimation techniques were applied for these purposes, respectively. Furthermore, this study discussed the relationship between causality and interpretability, and explore the definition framework of interpretability in traffic crash analytics. Causality is an important part of model interpretability, and model interpretability also includes four other aspects: trust, heterogeneity, transferability, and stability. This study can deepen researchers' understanding of model interpretability and facilitate the discovery of crash causation chains.

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2. Literature review

2.1. Influence factors

The potential factors influencing crashes can be categorized into five main groups: weather, temperature, traffic characteristics, temporal variables, and highway geometry.

Weather is a significant contributor to collisions and crashes (Rahman et al., 2011). While Alimo et al. (2023) found that most crashes occur under clear weather conditions, this does not imply that good weather is more likely to cause crashes than bad weather. Instead, it reflects the higher frequency of clear weather. Snowy conditions, for example, significantly increase the severity of heavy truck crashes (Ahmed et al., 2018), and snowy events have been identified as critical factors in crash studies in Iowa (Khattak and Knapp, 2001). However, weather does not always directly influence crashes. IM Almadi et al. (2023) highlighted that speeding and speed limits play a more significant role in crash occurrences during inclement weather. Reduced visibility due to weather can impair driver vision, lower road speeds, increase speed variability, and ultimately elevate crash risk (Ashley et al., 2015). Interestingly, not all studies associate adverse weather with increased crash risk. For example, rainy conditions were linked to a low possibility of crashes (Rusli et al., 2018).

Temperature is frequently considered in crash analysis, often treated as a component of weather (Jung et al., 2010). There was not a consistent conclusion about the effect of temperature on crashes (Theofilatos and Yannis, 2014). Brijs et al. (2008) found that lower temperatures (temperatures below 20 degrees) increase the number of crashes. Abnormal temperatures have been shown to significantly impact crash severity (Yazdani and Nassiri, 2021).

Large fluctuations of traffic characteristics can increase the traffic risk and contribute to crashes. Daily traffic volume has a non-linear relationship with crashes (Qin et al., 2004; Ivan, 2004). Crash frequency is affected by segment length (Qin et al., 2004) and traffic volume distribution throughout the day (Ivan, 2004). Monthly traffic volume has also been found to significantly increase traffic crash risk (Weng et al., 2021).

Considering that many factors are varying with time, the impact of temporal factors cannot be ignored. Wang et al. (2022) compared nighttime and daytime crashes, identifying distinct risk factors for each. Seasonality has also been used as a predictor in crash analysis (Jung et al., 2010).

Road geometry can affect the driving behavior and therefore influence the crashes. Factors including lane width (Musey and Park, 2016), combinations of horizontal and vertical curves (Jung et al., 2018), roadway curvature degree (Schneider et al., 2009; Musey and Park, 2016), and median guardrails (Mirbagheri et al., 2024) have been studied.

2.2. Model approach

Research on highway crash frequency has been conducted for many years. Lord and Mannering (2010) summarized potential data problems and listed effective methods for crash-frequency data analysis.

Traditional statistical models are widely applied, including generalized linear models, Poisson regression, survival theory, zero-inflated, negative binomial models, Zeger and Karim (1991), Miaou and Lum (1993), Jovanis and Chang (1989), Qin et al. (2004) and Milton and Mannering (1998). Additionally, various model structures have been proposed to solve problems of the raw data and parameter estimation procedure. For example, crash cases are relatively rare than normal cases, and many zero-inflated models were proposed, like the zero-inflated negative binomial model (Hosseinpour et al., 2014). Many studies have analyzed the temporal-spatial correlation, like the spatial autoregressive model, conditional auto-regressive model, and geographically weighted regression (Abdel-Aty and Wang, 2006; Ma and

Kockelman, 2006; Hosseinpour et al., 2018; Wang and Feng, 2019; Gaweesh et al., 2019).

ML techniques are being developed rapidly. ML models are strong at identifying the nonlinear relationship between variables, estimating model parameters unbiasedly, and reaching high-performance scores. Tree-based models, boosting techniques, deep neural networks, and other models have been applied (Yehia et al., 2021; Raja et al., 2023; Li et al., 2024). Chang and Chen (2005) utilized classification and regression tree (CART) to select the important factors in crash analysis, and CART performed better than traditional regression model.

BN have been applied in highway safety researches due to its simple and straightforward representation. Sun and Sun (2015) employed a dynamic BN to examine the relationship between crashes and speed. Yahaya et al. (2021) estimated factors to injury crashes on Ghanaian highways using BNs, combining with various oversampling techniques.

2.3. Bayesian network structure learning

The BNs structure learning techniques have three types: constraint-based, score-based, and hybrid algorithms (Kitson et al., 2023). Constraint-based methods rely on conditional independence tests to determine whether a connection exists between two variables, like the Peter and Clark (PC) algorithm (Spirtes and Glymour, 1991). Score-based methods, on the other hand, define a scoring function to evaluate model performance and search for the network structure that reaches the highest score. Prominent examples include the Hill Climbing algorithm and the Chow-Liu Trees algorithm (Heckerman et al., 1995; Chow and Liu, 1968). Hybrid algorithms combine both constraint-based and score-based approaches to take advantage of them.

A major challenge in structure learning is the computational complexity, as the process is NP-hard due to the large number of variables involved (Chickering, 1996; Chickering et al., 2004). However, in traffic safety analysis, the problem is relatively simpler because the number of potential factors is limited and manageable. Researchers in this field prioritize interpretability over computational efficiency, making it feasible to manually identify or modify BN structures.

Incorporating expert knowledge is a promising approach to enhance interpretability during structure learning. The integration of expert knowledge has evolved over the years and can be classified into soft and hard constraints, depending on whether they are mandatory during the learning process. For instance, Castelo and Siebes (2000) allowed experts to assign priors to arcs, indicating their presence or absence, even in spaces that include cycles larger than DAGs. Beyond arc-specific constraints, Borboudakis and Tsamardinos (2014) and Amirkhani et al. (2016) explored adding constraints on node ordering based on temporal and causal relationships. Li and Beek (2018) developed an interactive algorithm that enables researchers to participate in the structure learning process. In highway crash analysis, Leu and Chang (2013) incorporated expert knowledge to modify BN structures after converting fault trees into BNs. While post-hoc inclusion of expert knowledge can mitigate interpretability issues, it does not fundamentally resolve them. For example, reversing the direction of an arc may introduce loops, violating the DAG constraint.

3. Data preparation

3.1. Data description

This analysis utilized four primary datasets: (1) road information, (2) traffic volume data for 2022, (3) crash data for 2022, and (4) weather conditions. The study focuses on the HuNing Highway in Suzhou, China, a two-lane bidirectional highway spanning 14.176 kilometers. Ten cameras were strategically placed along the route to monitor hourly traffic volume. Based on intersections, the highway was divided into segments, with ten sections selected for analysis to align

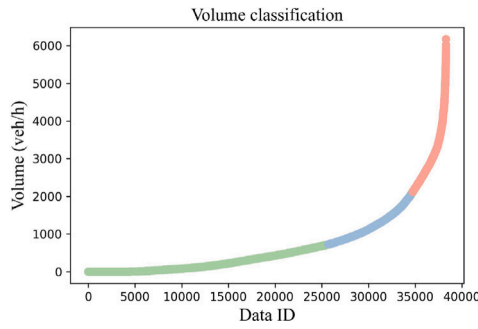


Fig. 1. The distribution of volume.

with the camera locations. Crash data was sourced from the Suzhou Road Traffic Crash Analysis and Warning System, which provided both statistical records and geographic visualization capabilities. The crash dataset includes detailed information on the time, location (longitude and latitude), and severity of each crash, enabling precise correlation with traffic volume data. Weather-related data, including time, temperature, and meteorological conditions, were obtained from www.tianqi.com.

3.2. Map matching

The four datasets were meticulously matched both temporally and spatially. Initially, ramps were used to divide the roadway into segments, ensuring consistent traffic volume within each section. Each segment was designed to be of nearly equal length, minimizing the influence of segment length on crash counts. Segments and cameras were grouped by traffic flow direction and matched as closely as possible within each group. The location descriptions of the cameras were then used to verify the corresponding outcomes. Segments without cameras were excluded from the analysis, and data related to the highway service area was also removed.

Crashes lacking time or location information were excluded from the database. After the cleaning process, the segments and crash data were aligned as precisely as possible. Crash location descriptions were crucial for ensuring accurate matching, and road segment IDs were used to link crash data with camera data. Ultimately, the dataset comprised 37,352 non-crash cases and 951 crash cases.

3.3. Data merging

Weather conditions for the relevant regions in 2022 were collected online, including both the highest and lowest temperatures. The linear relationship can be measured by the Pearson correlation coefficient r . The formula for r is:

$$r = \frac{\sum (x - \mu_x)(y - \mu_y)}{\sqrt{\sum (x - \mu_x)^2 \sum (y - \mu_y)^2}} \quad (1)$$

where x and y are two vectors, and μ_x and μ_y are their respective means. The correlation between the highest and lowest temperatures was calculated as 0.9480, indicating a strong positive relationship. Consequently, the mean temperature – the average of the highest and lowest temperatures – was computed and merged with the cases based on time.

The distribution of traffic volume is not uniform, as shown in Fig. 1. More than half of the data points are less than 1000 vehicle per hour (veh/h). In contrast, few data cases show much higher volumes. Considering the significant differences, it is necessary to classify traffic volume. Uijtdeuilligen et al. (2024) converted traffic volume into three categories: low, medium, and high. When estimating the effect of

Table 1

Description of variable for the analysis.

Variable	Description	Crash cases		Non-crash cases	
		Count	Percentage	Count	Percentage
Time of week	Weekday	664	69.84	25 894	69.32
	Weekend	216	22.71	8470	22.68
	Holiday	71	7.47	2988	8.0
Weather	Cloudy	580	60.99	23 114	61.88
	Sunny	124	13.04	5301	14.19
	Foggy	102	10.73	4277	11.45
	Rainy	99	10.41	4327	11.58
	Snowy	46	4.84	333	0.89
Volume	Low level	475	49.95	25 218	67.51
	Medium level	298	31.34	8645	23.14
	High level	232	24.4	3489	9.34
Temperature	Cold	421	44.27	12 174	32.59
	Cool	232	24.4	13 134	35.16
	Hot	298	31.34	12 044	32.24
Daytime	Dawn (3:00–7:00)	65	6.83	6303	16.87
	Morning (7:00–11:00)	182	19.14	6202	16.6
	Noon (11:00–15:00)	212	22.29	6181	16.55
	Afternoon (15:00–19:00)	204	21.45	6180	16.55
	Evening (19:00–23:00)	207	21.77	6199	16.6
	Night (23:00–3:00)	81	8.52	6287	16.83
Season	Spring	72	7.57	8309	22.25
	Summer	208	21.87	8275	22.15
	Autumn	267	28.08	9875	26.44
	Winter	404	42.48	10 893	29.16
Total		951		37 352	

weather conditions on crashes, Brijs et al. (2008) converted temperature into four classes: <0 , 0 – 10 , 10 – 20 , and >20 degrees centigrade ($^{\circ}\text{C}$). In this study, 0.62% of cases had mean temperatures below 0°C . Therefore, the mean temperature was divided into 3 categories. K-means clustering can group data based on the similarity, involving (Jain, 2010): (1) select initial cluster centers; (2) put each case to the nearest center; (3) compute new cluster centers as the centroid of each temporary cluster; (4) reassign observations to the closest cluster; and (5) repeat until convergence is achieved.

For this study, traffic volume data was categorized into three levels (high, medium, and low) using thresholds of 2117 and 724 (veh/h). Similarly, mean temperature was classified into three categories: cold, cool, and hot, with thresholds of 10.5 and 22.0°C , respectively. Daytime classification criteria were based on Zhou et al. (2021). Road geometry variables, such as lane width and curvature degree, were excluded because all segments shared identical geometric characteristics and were located in flat terrain. Descriptive statistics for the variables are summarized in Table 1. The dataset included segments with diverse properties observed over one-hour intervals, with segments experiencing crashes classified as crash cases and those without crashes as non-crash cases.

4. Methodology

BN is a graphical representation of causal relationships among variables, where nodes are variables and directed arcs show the causal relationships. The absence of arcs between nodes means that the variables are conditionally independent.

Let $D = \{X_1, X_2, \dots, X_n, n \geq 1\}$ be a set of variables. A BN is a DAG defined over D , accompanied by probability tables $X_p = [p(X_i|pa(X_i)), X_i \in X]$, where $pa(X_i)$ denotes the parent nodes of X_i . The BN represents the joint probability distribution of the variables as follows:

$$P(D) = \prod_{X_i \in X} P(X_i|pa(X_i)) \quad (2)$$

4.1. Structure learning

Building a DAG involves two key steps:

- Constructing a skeleton, which represents the undirected structure of the graph.
- Determining the directions of edges between nodes based on the skeleton.

This study focuses on combining expert knowledge with conditional independence tests to construct the Bayesian skeleton, thereby addressing the computational complexity and enhancing interpretability.

4.1.1. Skeleton selection

The skeleton $\mathcal{N}$ is an undirected graph obtained by removing the directions from all arcs in $\mathcal{N}$. The notation $I(A, B, S)$ indicates that node A is d -separated (where d denotes a directed graph) from node B given the set S . To incorporate and emphasize the importance of expert knowledge, this study introduces hard constraints based on expert knowledge and proposes the EKC algorithm for constructing the network structure. The process of building the skeleton of $\mathcal{N}$ consists of the following steps:

1. Based on expert knowledge (derived from relevant studies and common sense), determine whether an arc exists between two nodes. Three outcomes are possible: True, False, or Unknown.
2. For arcs labeled as Unknown, use a conditional independence test to determine their existence.

In step 1, “True” indicates that an arc exists between two nodes, “False” means no arc exists, and “Unknown” signifies uncertainty about the existence of an arc. In step 2, the non-parametric test, chi-square test, is employed to determine whether two categorical variables are independent. The formula for the chi-square test is:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad (3)$$

where χ^2 is the chi-square test statistic, $\sum$ is the summation operator, O is the observed frequency, and E is the expected frequency. In addition, there are many conditional independent tests for not only the categorized variables but also the continuous variables. Other non-parametric tests, and mutual information tests are alternative methods, such as Wilks’ G-square test, Mann–Whitney U test, and Fligner–Killeen test. For example, the Mann–Whitney U test and Fligner–Killeen test were used to identify the mean and variance difference between crash and non-crash cases, respectively, in real-time crash prediction (Wang et al., 2025). Considering that the results from different tests may be inconsistent, the Bonferroni correction is an option to solve the multiple errors (Hochberg, 1988; Abdi et al., 2007). The Bonferroni correction tests each hypothesis at $p = \alpha/m$, where p is the significance level, α is the desired overall alpha level, which is usually set as 0.05, and m is the number of hypotheses.

The skeleton selection criteria are summarized in Table 2. Specifying too many relationships between variables not only requires a great deal of prior expert knowledge, but also prevents the model from fully exploring the data itself. Therefore, when using judgment criteria, it is recommended that researchers focus more on setting in advance which relationships between variables are impossible.

4.1.2. Direction decision

After selecting the skeleton, the undirected graph requires the assignment of directions between nodes. To follow the fundamental constraints of a DAG, directed edges and acyclicity, the following steps are employed to determine the directions that are discussed in Jensen and Nielsen (2007):

Table 2

Skeleton selection criteria.

Expert knowledge	Independent test	Skeleton arc
True	True	True
True	False	True
False	True	False
False	False	False
Unknown	True	True
Unknown	False	False

1. Introduction of v-structure: If there exist a structure for three nodes like $A - B - C$ with no arcs between A and B , introduce v-structure $A \rightarrow C \leftarrow B$.
2. Avoid new v-structures: Once all v-structure have been introduced, no additional v-structures should be created. For example, if $A \rightarrow C - B$ exists after step 1, $C \rightarrow B$ must be applied and $B \rightarrow C$ is forbidden.
3. Avoid cycles: If assigning a direction $A \rightarrow B$ creates a cycle, revise the direction to ensure the graph remains acyclic.
4. Choose randomly: After applying steps 1–3, if any arcs remain undirected, assign them an arbitrary direction.

Step 4 can result in multiple potential networks due to the random assignment of directions. To identify the optimal network, score functions are utilized, as detailed in Section 4.4.

4.2. Parameter learning

After identifying BN structures, the interactions between variables are estimated using conditional probabilities. For complete data, maximum likelihood estimation (MLE) and Bayesian estimation are commonly employed. In this study, the Bayesian estimation method (Heckerman et al., 1995) was utilized to calculate the conditional probabilities due to the presence of a large proportion of zero values.

The distribution of θ , given the data $\{x_1, x_2, \dots, x_n\}$, is referred to as the posterior distribution, expressed as:

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{g(x)} \quad (4)$$

where $g(x)$ is the marginal distribution of X . The Bayes estimate of the parameter θ is the posterior mean when the loss function is defined as squared error. In the discrete case, the marginal distribution $g(x)$ can be calculated using the formula:

$$g(x) = \sum_{\theta} f(x|\theta)\pi(\theta) \quad (5)$$

where $f(x|\theta)$ is the likelihood function and $\pi(\theta)$ is the prior distribution of θ .

4.3. Inference

In highway crash analysis, the most dangerous scenario can be identified in the step of inference. BNs employ two primary approaches to probabilistic inference: exact inference and approximate inference. Given the relatively small sample size and limited number of variables in this study, the variable elimination algorithm, an exact inference technique, was selected for its efficiency in computing probabilities (Ghahramani and Jordan, 1993; Koller and Friedman, 2009).

The variable elimination algorithm can be summarized as follows:

- Construct factors: Define a factor (a function) for each node, representing a (conditional) joint probability.
- Restrict factors: Assign observed values to evidence variables (variables with known values) and restrict the factors accordingly.
- Eliminate hidden variables: Sequentially eliminate each hidden variable X_{h_j} by multiplying the remaining factors.
- Normalize factors: Normalize the resulting factors to convert them into a valid probability distribution.

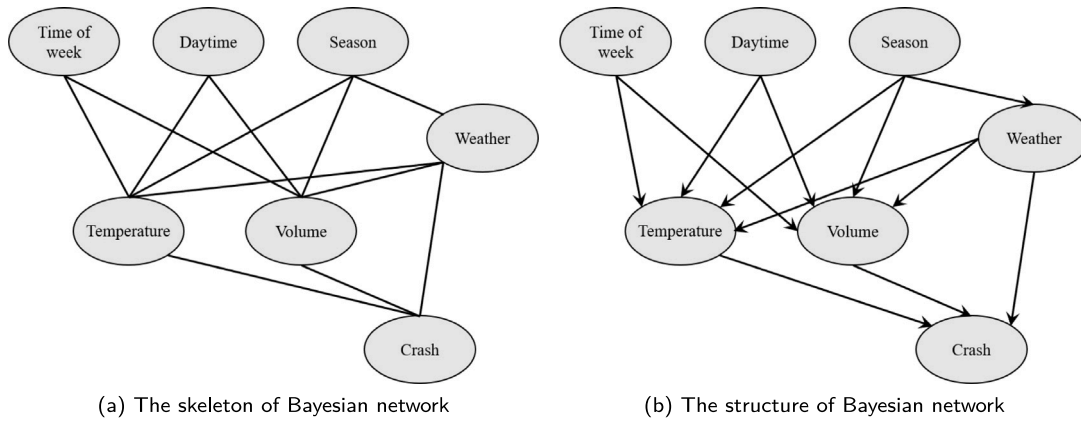


Fig. 2. The process of constructing the Bayesian network structure.

4.4. Score function

Four score functions, including K2, Bayesian Dirichlet equivalent uniform (BDeu), Bayesian Dirichlet sparse (BDs), and Bayesian Information Criterion (BIC). K2, BDeu, and BDs are developed from the general Bayesian Dirichlet (BD) score based on the assumption that the model parameters are Dirichlet prior for discrete data. BIC does not rely on any prior assumptions and has wider applications.

K2, BDeu, and BDs are belong to the BD family of scores. The basic function of BD is defined as:

$$S_{BD} = \prod_{i=1}^n \prod_{j=1}^{q_i} \left[\frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + N_{ij})} \prod_{k=1}^{r_i} \frac{\Gamma(\alpha_{ijk}) + N_{ijk}}{\Gamma(\alpha_{ijk})} \right] \quad (6)$$

where the indices i , j , and k represent the variable, parent node configuration, and variable state, respectively. q_i is the possible total number of combinations of values of the parents of the node X_i , and r_i is the possible values of node X_i . N_{ijk} denotes the count of samples in dataset D , and $N_{ij} = \sum_{k=1}^{r_i} N_{ijk}$. α_{ij} , the imaginary sample size, determines how much weight is assigned to the prior, and $\alpha_{ij} = \sum_{k=1}^{r_i} \alpha_{ijk}$.

- When $\alpha_{ijk} = 1$, the K2 score is obtained (Cooper and Herskovits, 1992).
- When $\alpha_{ijk} = 1/(r_i q_i)$, the BDeu is obtained (Buntine, 1991; Heckerman et al., 1995).
- When $\alpha_{ijk} = 1/(r_i \tilde{q}_i)$, the BDs is obtained (Scutari, 2016), where the $\tilde{q}_i$ is the presented number of combinations of values of the parents of the node X_i .

The BIC score, which is widely used for model selection, is defined as:

$$S_{BIC} = S_{LL}(DAG, D) - \frac{\log N}{2} \sum_{i=1}^n (r_i - 1) q_i \quad (7)$$

where $S_{LL}(DAG, D)$ is log-likelihood score and N is the sample size. The BIC score has been shown to facilitate faster learning of the true network structure compared to AIC and BDeu (Liu et al., 2012).

Above all, all four fitting functions can be applied to compare the model performance. BIC can be widely used to compare different models not only for the BN but also for the logistic regressions. K2 suits the scene with a large data scale and a simple model structure. BDeu can provide exact measurements for small data scales. BDs is appropriate for high-dimensional and sparse datasets. The smaller the BIC is, the better the performance is, but for the last three scoring functions, the larger the better.

5. Results

5.1. Structure learning

There are seven nodes in the network. If every pair of nodes is connected by an undirected arc, the total number of possible links would be $\binom{7}{2} = 21$. These connections are determined using both conditional independence tests and expert knowledge, as outlined in Table 2.

Expert knowledge suggests no positive correlations among certain variables. Specifically, three links, time of week – daytime, time of Week – season, and daytime – season, were removed, as these variables are derived from the timing of data recording and are not meaningfully related. Additionally, weather forecasting is a complex field requiring extensive data and high-performance computing systems (Alley et al., 2019), making it unlikely that weather is significantly associated with time of week or daytime. Consequently, five links were designated as False.

Certain connections are supported by prior research or logical reasoning. For example, the links between daytime – temperature and daytime – volume reflect historical patterns, such as higher temperatures during midday and peak traffic volumes during morning and evening hours. Similarly, the correlation between weather and temperature is well-documented, particularly in regions like Shanghai, which experiences hot summers, cold winters, and frequent summer rainfall. Empirical studies have also validated the relationships between weather – crash, temperature – crash, and volume – crash (Rahman et al., 2011; Jung et al., 2010; Qin et al., 2004). In total, seven links were confirmed as True. The remaining nine links, not addressed by expert knowledge, were evaluated using conditional independence tests.

Table 3 presents the results of the expert knowledge assessment and independent tests, illustrating the application of the skeleton selection criteria. The final network contains 21 connections, with six links (as shown in Table 3) where the results of the independent test and expert knowledge diverge. The independent test, relying solely on data correlations, may fail to capture the true significance of these relationships.

The summarized results are as follows:

- Expert knowledge test: Five links were determined to be False, seven links were confirmed as True, and nine links remained Unknown.
- Conditional independence test: Among the Unknown links, five were tested as True, and four were tested as False.
- Final result: Twelve links were identified in the network.

Table 3
Skeleton selection result.

Node A	Node B	Parent node	Expert knowledge	Independent test	Result
Time of week	Daytime	–	False	True	False
Time of week	Season	–	False	True	False
Time of week	Temperature	–	Unknown	True	True
Time of week	Weather	–	False	True	False
Time of week	Volume	–	Unknown	False	False
Time of week	Crash	–	Unknown	False	False
Daytime	Season	–	False	True	False
Daytime	Weather	–	False	False	False
Daytime	Temperature	–	True	True	True
Daytime	Volume	–	True	True	True
Daytime	Crash	–	Unknown	False	False
Season	Weather	–	True	True	True
Season	Temperature	–	True	True	True
Season	Volume	–	Unknown	True	True
Season	Crash	–	Unknown	False	False
Weather	Temperature	–	Unknown	True	True
Weather	Volume	Season	Unknown	True	True
Weather	Crash	Season	True	True	True
Temperature	Volume	Time of week	Unknown	True	True
		Daytime			
		Season			
		Weather			
Temperature	Crash	Time of week	True	False	True
		Daytime			
		Season			
		Weather			
Volume	Crash	Time of week	True	False	True
		Daytime			
		Season			
		Weather			

Noted: Bold-marks means the expert knowledge is different from the independent test results.

Fig. 2(a) illustrates the architectural representation of the network structure. Once the BN skeleton is determined, the directions of the links are assigned by introducing v-structures, as depicted in Fig. 2(b). This process ensures that all linkages are assigned directions without creating cycles, and the resulting architecture remains acyclic. Additionally, no undirected connections remain, eliminating the need for random direction assignments.

5.2. Parameter learning

To analyze the relationships among key factors influencing crash probabilities, this study employs a Bayesian estimation approach for computing conditional probabilities. The conditional probability distributions (CPDs) of significant variables are visualized using box plots, which effectively illustrate the impact of different parameter states. For instance, the temperature node, influenced by four parent nodes (weather, season, day of the week, and day), encompasses $3 \times 6 \times 4 \times 5 = 3600$ combinations of parent states. For each temperature condition, 3600 crash possibilities are evaluated, and their probability distributions are depicted through box plots.

The results of parameter learning reveal the CPDs of various variable states, with the median value representing the average effect of each state. The strength of the effect is indicated by the interquartile range (IQR), where a smaller IQR suggests a stronger influence from a specific factor, implying reduced contributions from other variables. Fig. 3 presents the CPD of crash cases, highlighting the impact of the three parent nodes (temperature, volume, and weather) through individual box plots. Further, Figs. 4 and 5 display the CPDs under low- and high-temperature conditions, respectively, while Figs. 6 and 7 illustrate the CPDs for medium-level volume and snowy conditions.

This approach provides a comprehensive understanding of how different factors contribute to crash probabilities, supported by clear visualizations that enhance the interpretation of results.

Fig. 3 presents the CPD related to crashes, highlighting key factors that significantly influence crash likelihood. Elevated crash probabilities are associated with extreme temperatures (both cold and hot),

medium-level traffic volume, and snowy weather conditions. High temperatures can increase tire pressure, while cold temperatures reduce road friction, both of which heighten crash risks. Adverse weather conditions, encompassing both temperature extremes, further exacerbate these risks. Statistically significant correlations have been established between crash severity in Iran and both low and high temperatures, with confidence intervals of 0.95 (Yahaya et al., 2021). Driving under such conditions necessitates reduced vehicle speeds and heightened driver concentration to mitigate risks.

Traffic volume also plays a critical role in crash probability. Medium-level traffic volume is associated with the highest likelihood of crashes within the specified range. Traffic congestion, characterized by a high volume of vehicles, and low-level conflicts, linked to fewer vehicles, both impact road safety. However, turbulence in traffic flow, which predominantly occurs in moderate traffic volumes, significantly compromises safety (Lee et al., 2003).

Weather-related factors further underscore the importance of environmental conditions in crash occurrences. Snowy weather, in particular, is associated with the highest crash likelihood, consistent with findings from prior studies (IM Almadi et al., 2023). Significant snow accumulation reduces road surface friction and driver visibility, making snowy and slippery conditions particularly dangerous. Research by Khattak and Knapp (2001) corroborates this, revealing a notable increase in crash rates during winter snow events compared to non-snow conditions on the same roadways.

In summary, extreme temperatures, medium traffic volume, and snowy weather are critical indicators of elevated crash probabilities. These findings emphasize the need for adaptive driving strategies and enhanced road safety measures under adverse conditions.

Fig. 4 illustrates the conditional probability distribution (CPD) for low-temperature cases, which predominantly occur during the winter season. Conversely, Fig. 5 displays the CPD for high-temperature cases, with the majority of such events observed in the summer. Notably, neither low nor high temperatures are significantly influenced by factors such as daytime, day of the week, or weather conditions in any

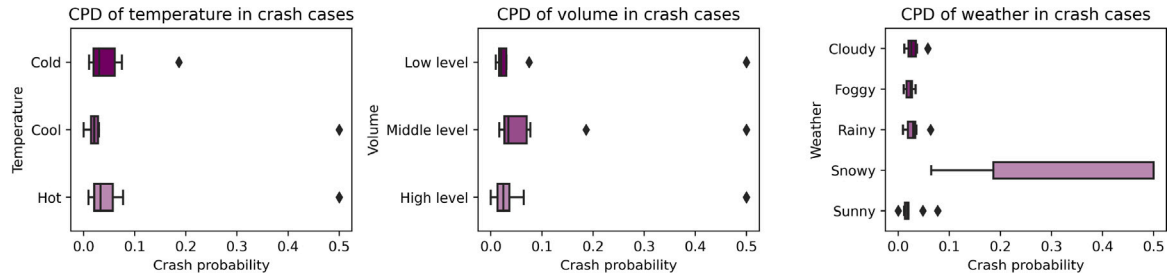


Fig. 3. The conditional distribution probability of crash cases.

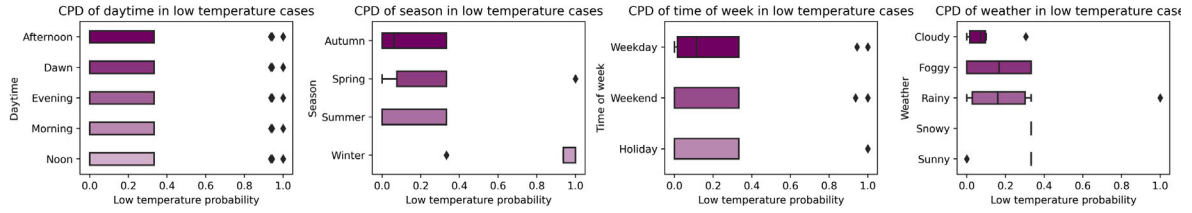


Fig. 4. The conditional distribution probability of low temperature.

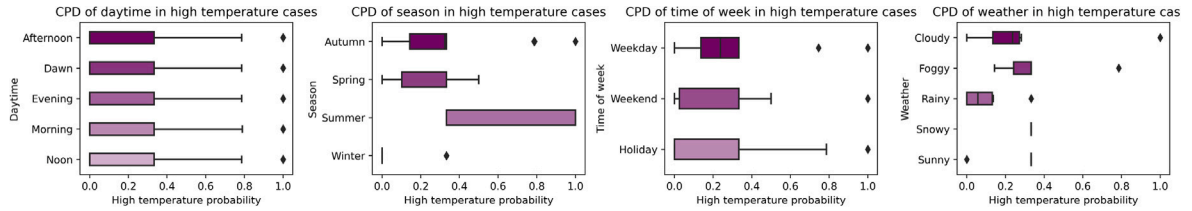


Fig. 5. The conditional distribution probability of high temperature.

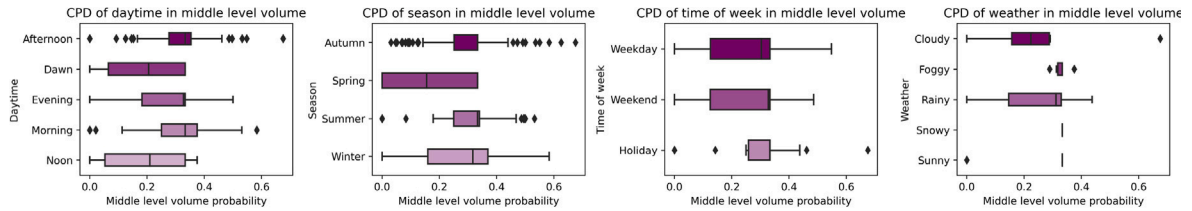


Fig. 6. The conditional distribution probability of medium volume.

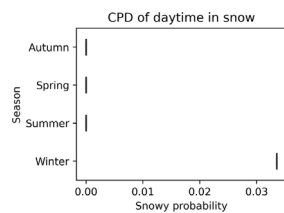


Fig. 7. The conditional distribution probability of snow.

given state. These findings highlight the seasonal dominance of temperature extremes, while underscoring the limited impact of temporal or weather-related variables on temperature variations.

Fig. 6 presents the CPD for medium traffic volume, which is most prevalent during peak traffic hours, specifically in the morning and afternoon. Winter exhibits a higher probability of medium traffic volume compared to other seasons. Additionally, holidays are more likely to maintain a stable medium volume effect than weekdays or weekends. Notably, overcast and wet weather conditions show a wide interquartile range, indicating greater variability in their impact on traffic volume.

Fig. 7 depicts the CPD for snowy conditions, which align with common expectations, as winter has the highest probability of snow-fall. These findings highlight the influence of temporal, seasonal, and weather-related factors on traffic volume and road conditions.

5.3. Inference

Inference in this study integrates all potential variables to identify high-risk scenarios. Among the seven nodes in the BN, six serve as potential influence variables, excluding the crash node. This results in a total of $3 \times 6 \times 4 \times 3 \times 3 \times 5 = 3240$ possible scenarios. Using a variable elimination approach, the crash probability for each scenario was calculated. The ten most dangerous scenarios, ranked by crash probability, are presented in Table 4.

Crash probability refers to the likelihood of a specific crash scenario occurring among all possible crash scenarios, with the sum of probabilities across all scenarios equaling 1. Table 4 highlights the top 10 most dangerous scenarios out of the 3240 possibilities. The most dangerous conditions occur on weekday mornings during winter, characterized by medium traffic volume, low temperatures, and cloudy weather. These findings align with the outcomes of parameter learning: winter and low temperatures signify poor driving conditions, cloudy

Table 4
The most ten dangerous situation.

ID	Volume	Temperature	Season	Weather	Daytime	Time of week	Crash probability
1	Medium	Cold	Winter	Cloudy	Morning	Weekday	4.81e-4
2	Medium	Cold	Winter	Cloudy	Noon	Weekday	4.71e-4
3	Low	Cold	Winter	Cloudy	Night	Weekday	4.38e-4
4	Low	Cold	Winter	Cloudy	Dawn	Weekday	4.22e-4
5	Medium	Cold	Winter	Cloudy	Afternoon	Weekday	4.07e-4
6	Medium	Cold	Winter	Cloudy	Evening	Weekday	3.59e-4
7	Medium	Hot	Summer	Cloudy	Afternoon	Weekday	3.29e-4
8	Low	Cold	Winter	Cloudy	Noon	Weekday	3.21e-4
9	Medium	Hot	Summer	Cloudy	Noon	Weekday	2.75e-4
10	Low	Cold	Winter	Cloudy	Afternoon	Weekday	2.29e-4

weather reduces visibility, weekday mornings represent peak traffic hours, and medium traffic volume is associated with higher conflict rates compared to low or high volumes. This analysis underscores the critical factors contributing to crash risks and provides actionable insights for improving road safety.

5.4. Model comparison

5.4.1. The compared model

This study conducted a comparative analysis of three BN structure learning algorithms: the proposed EKC algorithm, Hill Climbing, and Chow-Liu Trees, with logistic regression serving as the benchmark classification model. The computational cost of traffic crash analysis scene is low. This study involves 7 variables, and the total number of possible links is 21. Compared to the techniques with complex mechanisms, the Hill Climbing and Chow-Liu Trees algorithms not only get good results but are also easily understandable.

The Hill Climbing algorithm operates as a heuristic optimization method that incrementally improves solutions through local maximization of objective score function. As a local search technique, Hill Climbing evaluates neighboring states at each iteration and moves to the state with the highest objective value, terminating when no better neighbors exist. It is prone to converge to local optima due to its greedy nature. In contrast, the Chow-Liu Trees algorithm is a constrained structural approach. It constructs a maximum likelihood tree-structured BN by limiting each node to at most one parent node. The algorithm employs mutual information between node pairs as edge weights and solves the maximum spanning tree problem to identify the optimal structure. For this study, we designated “time of week” as the root node in Chow-Liu Trees implementations to reflect temporal dependencies as the foundational hierarchy in the network. This constraint not only reduces computational complexity from exponential to polynomial time but also aligns with real-world scenarios where temporal factors often dominate causal relationships.

The modeling environment is Python, and the Python library for BN construction is mainly based on pgmpy (Ankan and Textor, 2024). The BN structure was constructed manually combined with the expert knowledge. 951 crash cases and 37,352 non-crash cases were all applied for scoring BN structure, learning parameter and inferring dangerous scene. The parameter learning technique and inference technique were applied with default settings. The code is available at github.com/yifanwang1017/EKC.

As a non-graphical model baseline, logistic regression was included to contextualize the performance evaluation. The dependent variable is a binary variable, including crash and non-crash. The distributions of crash variable and other variables are shown in Table 1. The prior distribution of parameters in the logistic model follows the default normal distribution. All models were systematically compared through rigorous model fitting assessments, with particular emphasis on evaluating the novel EKC algorithm’s effectiveness relative to established methodologies.

5.4.2. Comparison results

A comparative analysis of BN structures learned through Hill Climbing and Chow-Liu Trees algorithms reveals notable structural discrepancies, as illustrated in Fig. 8. Notably, the Hill Climbing algorithm-derived structure (Fig. 8(a)) contains a counterintuitive dependency between season $\rightarrow$ time of week, while the Chow-Liu Trees algorithm-generated model (Fig. 8(b)) presents an equally problematic weather $\rightarrow$ season linkage. These empirically inconsistent connections underscore the limitations of conventional structure learning approaches in maintaining domain coherence. In contrast, the proposed EKC algorithm demonstrates superior interpretability by systematically eliminating such paradoxical dependencies through explicit domain knowledge integration.

Table 5 presents comparative scoring metrics for BNs and logistic regression models, with scoring functions formally defined in Section 4.4. The proposed EKC algorithm demonstrates optimal performance in information-theoretic criteria, achieving minimal BIC scores ($-243,757.28$). While the Hill Climbing algorithm attains superior K2 ($-235,771.14$), BDeu ($-235,556.36$), and BDs ($-235,620.01$) scores, these results must be interpreted in light of methodological constraints.

The Hill Climbing algorithm’s greedy search mechanism predisposes it to local optima convergence, as evidenced by structural instability across multiple runs. Chow-Liu Trees’ restrictive single-parent architecture fundamentally limits its capacity to capture multivariate dependencies, resulting in oversimplified network topologies.

To identify whether the scores’ differences were significant or not, a newly managed dataset was used. The time of week, volume, and crash were chosen, and the entire dataset was applied. The simplified BN was constructed with only two links, time of week $\rightarrow$ volume and volume $\rightarrow$ crash. The BIC of the simplified BN is $-66,572.84$ and is significantly smaller than the BIC numbers of three completed BN, which means the score difference between various algorithms shown in Table 5 was small.

In this study, the proposed EKC algorithm can improve the model interpretability in causal relationship discovery by adding expert knowledge. The improvement of trust, heterogeneity, transferability, and stability requires further efforts and will be in future directions.

6. Discussion

This study primarily focuses on the causal relationship and aims to enhance model interpretability. However, causality is not equal to interpretability. Furthermore, the meaning and scope of model interpretability in traffic crash analytics are not summarized. This section further discuss model interpretability, including causality, trust, heterogeneity, transferability, and stability.

Model interpretability does not address issues arising from traffic data itself or data processing methods, such as selection bias due to non-representative crash reporting (Hauer, 2006). Interpretability levels vary depending on the expertise of the audience. Interpretability is generally intended for researchers with basic expertise, not those entirely unfamiliar with traffic crash analytics. In ML, interpretability lacks a consistent definition. One notion of interpretability is “you

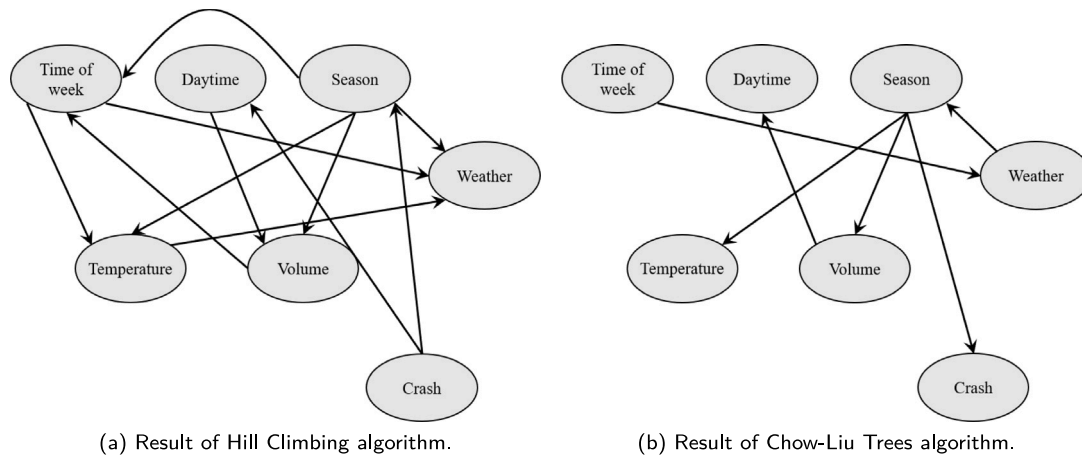


Fig. 8. The Bayesian network structure results of Hill Climbing algorithm and Chou-Liu Trees algorithm.

Table 5

The fitting results comparison between different models.

Model	BIC	K2	BDeu	BDs
EKC algorithm (our model)	-243,757.28^a	-238,913.00	-239,081.37	-240,110.75
Hill climbing algorithm	-236,221.37	-235,771.14	-235,556.36	-235,620.01
Chow-Liu trees algorithm	-239,643.44	-239,602.29	-239,596.56	-239,615.28
Logistic model	8656.33			

^a The bold-mark is the best performance among models.

will know it when you see it”. The basic definition requires detailed explanations in different areas. Zhang et al. (2021) emphasize the need for explanations and understandable terms, while Doshi-Velez and Kim (2017) thought that “interpretability aids in understanding fairness, privacy, reliability, robustness, causality, usability, and trust”. For example, in computer vision (CV), interpretability is often achieved through visual outputs like superpixels or visual concepts (Zhang et al., 2021). It is not easy to strictly compare the interpretability level between traditional statistics and ML models, but some ideas are widely accepted. Linear models are not inherently more interpretable than deep neural networks because linear models are limited in extracting high-dimensional features (Lipton, 2018).

Causality refers to a relationship where the cause partially influences the outcome, and the outcome partially depends on the cause. Correlation indicates that two variables exhibit a similar trend (Altman and Krzywinski, 2015), but it does not imply causality. While data-driven models may reveal strong relationships that pass scientific scrutiny, such relationships do not always reflect causal attributes. Causality is critical, as emphasized by Hauer and Jovanis (2020): “Effective prevention rests on knowledge of causes”. Causal relationships between variables involve both causal direction and causal effect (Yao et al., 2021). For instance, Wallén Warner et al. (2008) illustrated crash causal relationships, linking causes to crash events, while Hauer (2020) further explored the significance of crash causal chains and their implications for preventive measures. In traffic crash analysis, the treatment effects of various interventions, such as work zones (Zhang et al., 2022a,b, 2023) and speed cameras (Li et al., 2013; Li and Graham, 2016), have been extensively studied.

Trust can be estimated objectively and subjectively. The quantitative metrics are important for evaluating crash model performance, like recall in real-time crash prediction (Wang et al., 2025). Relying on quantitative measures, however, can lead to an overemphasis on high scores and an ignorance of the crash causal chain. Models with large research histories and widespread applications often inspire greater confidence. Studies showed that engineers may prioritize experience-based policies even when evidence contradicts their intuition (Davis, 2012; Hauer, 2019), highlighting the intrinsic link between trust and

interpretability. Additionally, ethical considerations and privacy concerns are essential subjective dimensions of trust. Haghighattalab et al. (2019) found that ethical considerations are often overlooked in crash analysis models.

Unobserved heterogeneity means that the estimation effect of a potential crash variable may be affected by the other variables (Mannering and Bhat, 2014). Unobserved heterogeneity can result in biased parameter estimates and erroneous conclusions. Mannering et al. (2016) discussed the unobserved heterogeneity in both variable and methodological aspects. Many methods have developed, and the random parameter technique is widely applied (Yu and Abdel-Aty, 2013; Dzinyela et al., 2024).

Many regions lack the necessary data, funding, and facilities, resulting in insufficient data availability across traffic networks. A model with high transferability can significantly reduce investments in traffic safety and eliminate redundant analysis processes. Transfer learning techniques have been successfully developed in many study areas, such as short-term traffic prediction (Li et al., 2021) and real-time crash risk prediction (Shew et al., 2013; Xu et al., 2014; Man et al., 2022).

Crash frequencies and time-related factors often exhibit instability over months or years. A notable challenge is regression-to-the-mean, a phenomenon where extreme crash counts at a site naturally revert to their long-term average. For instance, traffic treatments are typically implemented at sites with high crash counts. Yet, due to the inherently low probability of crash occurrence, the observed reduction in crashes may reflect a return to the mean rather than the effectiveness of the interventions. Thus, attributing crash reductions solely to traffic treatments can be misleading. Analyzing temporal stability is crucial for accurately assessing treatment effects. Mannering (2018) explored methods to address temporal stability and enhance interpretability. The likelihood ratio test can evaluate the temporal stability of model estimates across various scenarios (Alogaili and Mannering, 2022).

7. Conclusions

This study makes two primary contributions: (1) developing the semi-data-driven Expert Knowledge-Constrained (EKC) algorithm that synergizes data-driven learning with domain expertise through a priori

causal constraints, and (2) demonstrating its superior performance through comparative analysis with established benchmarks. Furthermore, this study discussed about relationship between causality and interpretability, The definition framework of interpretability includes causality, trust, heterogeneity, transferability, and stability.

A multi-source dataset from HuNing Highway in 2022 was used in this study, integrating crash records, roadway geometry, meteorological conditions, and traffic operations through spatiotemporal matching. The EKC algorithm based BN operationalizes interpretability through three phases:

- **Structure learning:** Implements hard causal constraints derived from traffic safety literature or common sense prior to network initialization, effectively pruning counterfactual dependencies.
- **Parameter learning:** Employs Bayesian estimation to compute conditional probability distribution tables under incomplete data conditions.
- **Inference:** Utilizes variable elimination to quantify crash risk through posterior probability maximization.

Comparative validation against Hill Climbing, Chow-Liu Trees, and logistic regression reveals that EKC achieves competitive goodness-of-fit while maintaining causal transparency. Key insights include:

- **Causal relationships:** Direct crash predictors emerge as temperature, traffic volume, and adverse weather, while temporal factors (season, time of week) exhibit only mediated effects.
- **Risk amplification:** Extreme temperatures, middle-level traffic volume, and snowy conditions increase crash likelihood compared to other ranges.
- **High-Risk scenario:** The configuration medium volume, cold temperature, winter, cloudy, weekday mornings has the largest crash proportion.

The EKC Bayesian network help to apply safety interventions, like prioritizing anti-icing measures on segments with historical cold-weather crashes, implementing adaptive speed control, and developing weather-responsive traffic management protocols. The definition framework of model interpretability provides a standard for comparing models. Researchers can systematically improve model interpretability when designing model algorithm.

While demonstrating methodological advancement, this study faces many constraints. Single-freeway analysis (HuNing Highway) limits transferability to divergent roadway types. Crash-to-non-crash ratio may bias probability estimates. The EKC algorithm's reliance on expert-driven causal constraints introduces non-trivial computational overhead compared to purely data-driven approaches. Future work should integrate real-time speed data and test the framework on urban road networks. The EKC algorithm's modular design permits adaptation to other transportation safety challenges, including pedestrian risk analysis and freight corridor optimization. As a probabilistic model, another potential future research direction is uncertainty quantification, which can improve model performance and reliability.

CRedit authorship contribution statement

Yifan Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Xuesong Wang:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no conflicts of interest to this work.

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